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Molecules Meet Language: Confound-Aware Representation Learning and Chemical Property Steering in Transformer-VAE Latent Spaces



(/pdf?id=WUx3s3ekOg)

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04 May 2026 (modified: 28 May 2026) NeurIPS 2026 Conference Submission

Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors

Revisions (/revisions?id=WUx3s3ekOg)

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Abstract:

Molecular generative models often assume meaningful latent geometry, but apparent property predictability can reflect sequence-level shortcuts rather than chemical organization. We study this issue in an unsupervised autoregressive Transformer-VAE trained on SELFIES. After training, we freeze the model, fit linear probes to RDKit descriptors, and use the probe weights as candidate global steering directions. To separate chemical signal from SELFIES artifacts, we introduce a confound-aware evaluation based on residualization, confound-direction alignment analysis, and decoded-molecule traversal. This is necessary because SELFIES length, branch tokens, ring tokens, and token entropy are strongly encoded in the latent space. Under this confound-aware evaluation, we find robust monotonic steering for cLogP, FractionCSP3, HeavyAtomCount, TPSA, BertzCT, and HBA. Nonlinear probes further show that some properties admit stable global directions, while others are better described by local latent gradients. Overall, our results show that chemically meaningful steering can emerge in entangled molecular latent spaces, but only when validated through decoded molecules and controlled for representation-level confounds.

Checklist Confirmation: I confirm that I have included a paper checklist in the paper PDF.

Responsible Reviewing: We acknowledge the responsible reviewing obligations as authors.

Primary Area: AI/ML for physical sciences (e.g., physics, chemistry, Earth sciences, astronomy)

Secondary Area: Deep learning advancements (e.g., architectures, optimizers, representation learning)

Contribution Type: General: Most submissions will fall into this type.

Academic Integrity: I acknowledge that I have read the NeurIPS Handbook and commit to adhering to all policies in the Handbook (<https://neurips.cc/Conferences/2026/MainTrackHandbook> (<https://neurips.cc/Conferences/2026/MainTrackHandbook>)), the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy.

LLM Usage: Prefer not to declare (you can then skip the rest)

Declaration: I confirm that the above information is accurate.

Reviewer Nomination: Zakaria Elabid (/profile?id=~Zakaria_Elabid1)

Submission Number: 24784

Discussion (?id=WUx3s3ekOg#discussion)

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Submission24784...



Add:

Confidential concern regarding reviewer up8r's final assessment 1/2

Author AC Confidential Comment

by Authors (Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

03 Aug 2026, 12:27 Program Chairs, Senior Area Chairs, Area Chairs, Authors

Comment:

Dear Area Chair,

We wish to raise a serious concern regarding the technical competence and reliability of reviewer up8r's final assessment. This is not a complaint that the reviewer remained negative. We accept that a reviewer may consider an evaluation framework based on computed molecular descriptors insufficiently original or significant for NeurIPS. Our concern is that the final review contains multiple fundamental misunderstandings of the submitted architecture, objective, and methodology, while assigning confidence 5 and lowering the rating after the requested evidence was supplied.

The reviewer reframes the paper as an attempt to "use LLM to learn chemical space" and asks why we do not "use transformer directly." The paper does not propose an LLM, and the submitted model is already an autoregressive Transformer-VAE, with a Transformer encoder and Transformer decoder. The VAE is not an alternative to the Transformer: its continuous bottleneck is the object being studied and is required to define the decoded trajectory ($z + \alpha d$). Removing it would remove the latent geometry under investigation.

The reviewer further claims that supervised fine-tuning on the properties could "do the same thing." This is technically incorrect. Supervised fine-tuning would inject the target properties and modify the representation. Our question is whether property organization emerges after property-blind generative training, before property supervision, and whether it remains readable and actionable with the generator frozen. A supervised prediction experiment therefore answers a different question.

The statement that linear probing is "out-dated" similarly misunderstands its methodological role. Linear probing is not used as a state-of-the-art predictor. Its coefficient vector defines one constant global direction, whose decoded actionability can then be tested. A more expressive predictor may recover more information but does not establish the existence of a reusable global axis. The manuscript explicitly uses MLPs only to diagnose additional nonlinear recoverability and decoded traversal as the operational steering test.

More seriously, the reviewer states that latent spaces of modern deep-learning models are "likely incompatible with ML models." We do not know of a coherent technical meaning for this assertion. Neural latent vectors are numerical feature representations, and fitting restricted estimators to frozen neural representations is a standard means of analysing their information and geometry.

The baseline requests exhibit the same task mismatch. Encoder-oriented or supervised property models such as Uni-Mol or ChemBERTa do not automatically provide a continuous generative latent space and a decoder applicable after displacement. ChemSpacE was selected because it performs the same relevant operation: post-hoc learning of a global direction in a frozen generative latent space followed by decoded traversal. We implemented it using the same representation, decoder, seeds, candidate budget, direction normalization, and displacement. The reviewer identifies no error in this protocol or its results and dismisses it only as "another weak model."

The SELFIES discussion also disregards the supplied evidence. Each canonical molecule contributes one deterministic SELFIES sequence; randomized aliases are not mixed during training, and generated outputs are decoded and canonicalized before evaluation. More importantly, the matched SMILES model achieved 97.65% reconstruction and strong residual descriptor recovery but only 9.33% traversal validity, compared with 100% for

SELFIES. More than 90% of displaced SMILES points were invalid. Describing this as a “slight error” is incompatible with the operational evaluation used in the paper. The additional assertion that SELFIES fails around 100 atoms was unsupported, uncited, and unrelated to the molecular regime or claims studied here.

The reviewer also dismisses HeavyAtomCount as trivial, although it is deliberately included as a size-sensitive control, not as a challenging property-prediction benchmark. Likewise, the original statement that cLogP might be explained by molecules becoming “more hydrophobic” confuses the target property, lipophilicity with a confound. We nevertheless addressed the technically meaningful concern, namely mediation by size or carbon-chain growth, through residual traversal and explicit molecular-change decomposition.

Most importantly, after we completed residual-direction traversal, nonlinear confound modelling, graph and functional-group decomposition, a matched SMILES experiment, the controlled ChemSpacE comparison, and fingerprint-constrained molecular design, the reviewer identified no implementation error, statistical error, unfair comparison, or incorrect numerical claim. Instead, the review shifted to demanding a supervised property-prediction study and repeated the architectural and methodological misunderstandings above.

Confidential concern regarding reviewer up8r's final assessment 2/2

Author AC Confidential Comment

by Authors (👁 Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

📅 03 Aug 2026, 12:28 👁 Program Chairs, Senior Area Chairs, Area Chairs, Authors

Comment:

The revision history shows that the rating was reduced from 2 to 1, Quality, Clarity, and Originality were reduced to 1, and confidence was increased from 4 to 5 after the requested experiments were supplied. Confidence 5 asserts absolute certainty, close familiarity with the literature, and careful checking of the technical details. That declaration is difficult to reconcile with failing to recognize that the model already uses a Transformer, treating Supervised FineTuning as equivalent to unsupervised representation analysis, claiming that neural latent spaces are incompatible with ML estimators, and characterizing a 9.33%-versus-100% validity result as a slight difference.

We therefore respectfully request that this score not be given normal evidentiary weight in the final decision. The issue is not merely disagreement over significance; the final assessment is based on repeated factual and methodological mischaracterizations and does not constitute a reliable technical evaluation of either the submitted paper or the evidence added during discussion.

Official Review of Submission24784 by Reviewer ysy4

Official Review by Reviewer ysy4 📅 10 Jul 2026, 20:57 (modified: 04 Aug 2026, 15:46)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer ysy4

📄 Revisions (/revisions?id=egoWykWAiw)

Summary:

The authors systematically study whether self-supervised autoregressive Transformers trained on molecular SELFIES strings implicitly learn latent spaces that permit property-guided molecular steering and generation. The authors' conclusions, in short, seem to suggest the answer is positive. Additionally, the authors study whether (and, if so, which) molecular properties are globally linear in such latent spaces or are nonlinearly local in nature.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Points of Strength:

1. From a machine learning perspective, as well as that of a chemical domain scientist, the authors' constructed study is interesting and meaningful. Showing that self-supervised autoregressive Transformers learn semantically informative (and generatively steerable) latent spaces for certain data domains is a useful finding for future research at the intersection of generative modeling and representation learning.
2. The authors' writing is relatively clear and easy to follow.
3. The authors' provided source code is well-documented and appears to be reproducible.

Points for Improvement:

1. Figure 1 is hard to read. For instance, the font size of this figure is very small (n.b., I have to zoom in considerably to read it fully). Also, some of the phrases included in the figure (e.g., "Steer along w , *monotonic latent traversal*) seem awkward and perhaps LLM-generated. I recommend the authors revise this figure to make it much easier for readers to digest when quickly glancing at it for a summary of this work.
2. The statement "This makes the method efficient and interpretable, but also imposes a stricter validation burden" in Lines 109-111 seems redundant by this point in the paper. By this section of the paper, the authors have already explained several times (in the abstract, introduction, etc.) that one should be careful about conflating linearly steerable properties and molecular (SELFIES) confounders.
3. The authors should reconsider their definition of molecular novelty during generative modeling. I recommend that the authors instead consider using a tool like FPSim2 (<https://github.com/chembl/FPSim2>) to efficiently compute molecular fingerprints for generated molecules and set a numerical threshold for what determines whether a generated molecule is sufficiently dissimilar from all the fingerprints of the training dataset to be considered novel (e.g., being below a threshold of 0.7 Tanimoto similarity might work, but the authors should do a literature review before settling on this value).

Quality: 3: good

Clarity: 3: good

Significance: 3: good

Originality: 3: good

Questions:

- Could the authors revise Figure 1 and their definition of molecular novelty? Doing so would likely raise my score.

Limitations:

The authors have addressed the limitations of this work.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

I did not find any major paper formatting issues.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

Rebuttal by Authors

Rebuttal

by Authors ( Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

 28 Jul 2026, 12:48 (modified: 28 Jul 2026, 16:08)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

 Revisions (/revisions?id=tKrzMH2YQP)

Rebuttal:

We thank the reviewer ysy4 for the positive assessment and for identifying three concrete improvements. We have addressed the principal requests as follows:

- Redundant Sentence: We will remove the redundant sentence noted around Lines 109–111 in the revision.
- Figure 1: We completely redesigned Figure 1 to improve readability at normal manuscript scale. The revised figure uses substantially larger text, removes unnecessary legends and clutter, and more clearly separates the model-training, probing, confound-control, and decoded-traversal stages. We also replaced the awkward traversal wording highlighted by the reviewer with concise technical notation. We apologize if it gave the impression that it was LLM generated, and as such both the revised figure and its editable drawio.xml source are provided in `rebuttal/E01_figure1/` in the anonymous repo from the paper.
- Fingerprint-based molecular novelty: We agree that exact canonical-string mismatch is insufficient as the primary novelty criterion. We therefore recomputed novelty using FPSim2 against the training partition only, with Morgan fingerprints of radius 2 and 2,048 bits. All 5,000/5,000

generated samples were valid, yielding 4,983 distinct canonical structures; the difference of 17 reflects duplicates rather than invalid generations.

Using the state of the art 0.70 maximum-Tanimoto threshold, 4,971/4,983, or 99.7592, of the distinct generated structures are novel relative to the training set. Sensitivity analyses give 99.8395 at 0.80 and 99.8595 at 0.90. The median nearest-training similarity is 0.2373, with 95th and 99th percentiles of 0.3997 and 0.5517, respectively. We use 0.70 as the primary reviewer-requested operating threshold, while explicitly avoiding the claim that it constitutes a universal definition of chemical novelty. Full scripts, tables, distributions, and provenance are available in [rebutta1/E02_fpsim2_novelty/](#) in the anonymous repo from the paper.

Anonymous Repo Link: <https://anonymous.4open.science/r/MolecuLA-7A68/>
(<https://anonymous.4open.science/r/MolecuLA-7A68/>)

Response to the authors' rebuttal

Official Comment by Reviewer ysy4 📅 29 Jul 2026, 01:13

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Thank you for the precise rebuttals. With the authors' latest responses, my primary concerns have been addressed.

➡ *Replying to Response to the authors' rebuttal*

Official Comment by Authors

Official Comment

by Authors (👁 Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

📅 03 Aug 2026, 16:48

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Dear Reviewer ysy4,

Thank you for confirming that your primary concerns have been addressed. Since you previously indicated that resolving the Figure 1 and fingerprint-based novelty issues would likely raise your score, we would be grateful if you could update the rating accordingly, should you still consider that appropriate.

➡ *Replying to Official Comment by Authors*

Response to Comment by the Authors

Official Comment by Reviewer ysy4 📅 03 Aug 2026, 19:19

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Thank you for addressing these concerns. At this time, I am comfortable with my current score of 5 for this paper.

Official Review of Submission24784 by Reviewer up8r

Official Review by Reviewer up8r 📅 24 Jun 2026, 08:38 (modified: 04 Aug 2026, 15:46)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer up8r

📄 Revisions (/revisions?id=r5xIhzNxEU)

Summary:

The paper studies whether an unsupervised SELFIES Transformer-VAE learns latent directions related to molecular descriptors. The authors freeze the VAE, fit linear probes to RDKit properties, use probe weights as steering directions, and evaluate decoded molecules along these directions with additional confound analyses.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths.

The proposed problem is meaningful. How to transverse in latent space is important for molecule representation and optimization tasks.

Weaknesses.

The evidence for chemically meaningful steering is not convincing. Take the cLogP experiment as an example. Increasing cLogP along a latent direction may simply mean that the decoder is producing larger or more hydrophobic molecules, for example by extending aliphatic chains. This would be a fairly trivial effect rather than evidence of a useful optimization direction. The paper controls for a few SELFIES-level statistics, but it does not control for more chemically relevant factors such as carbon count, scaffold, or functional groups. Also, as far as I can tell, the traversal experiments use raw property directions rather than residualized or confound orthogonalized directions. Therefore, the observed monotonic increase in cLogP does not rule out a simple size or lipophilicity confound.

I also find the MLP probing analysis hard to understand. The MLP probes show that some descriptors can be nonlinearly predicted from the latent representation, but this is merely a diagnostic on its own. Nonlinear decodability does not mean that the latent space supports useful transversing, nor does it learn a chemically meaningful geometry. A finetuned predictor, or even a standard molecular fingerprint baseline, could likely achieve similar descriptor prediction. If the paper's goal is to understand the latent space, it needs evidence that the representation supports controlled molecular edits or optimization, not just descriptor prediction. Like conducting a seed-to-lead molecule optimization experiments, directly compared with: CLaSMO (<https://arxiv.org/abs/2411.01423> (<https://arxiv.org/abs/2411.01423>)), GBGA (<https://pubs.rsc.org/en/content/articlelanding/2019/sc/c8sc05372c> (<https://pubs.rsc.org/en/content/articlelanding/2019/sc/c8sc05372c>)), or MARS (<https://arxiv.org/pdf/2103.10432> (<https://arxiv.org/pdf/2103.10432>)).

The lack of external baselines is another issue. Most comparisons are between variants of the authors' own VAE architecture. This makes it difficult to tell whether the reported behavior is specific to the proposed Transformer-VAE, or whether it is a generic consequence of molecular string representations and descriptor correlations. As I can tell, the SELFISH is not a suitable representation for such tasks, for the same molecules will have many different SELFISH. Additionally, SELFISH is weaker than SMILES when tackling bigger molecules.

More broadly, the problem formulation is somewhat ambiguous. If the paper is meant as a representation-analysis study, then they should compare with more baselines, such as UniMol, MolAE, Suiren. Based on that, they should also propose useful trick or advice on improving Molecular FM's latent space. If the paper is meant to demonstrate useful molecular design method, then the current experiments are apparently not enough. A more convincing evaluation would include seed-to-lead or constrained molecular optimization, with scaffold or fingerprint similarity constraints, and comparisons against existing molecular optimization methods.

Quality: 1: poor

Clarity: 1: poor

Significance: 3: good

Originality: 1: poor

Questions:

See weaknesses part.

Limitations:

See weaknesses part.

Confidence: 5: You are absolutely certain about your assessment. You are very familiar with the related work and checked the math/other details carefully.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

The result tables are hard to read.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

Rebuttal by Authors

Rebuttal

by Authors (👁 Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

📅 28 Jul 2026, 13:35 (modified: 28 Jul 2026, 16:08)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

📄 Revisions (/revisions?id=tQ7hssWKpU)

Rebuttal:

We thank reviewer up8r for recognising the importance of latent-space traversal for molecular representation and optimisation. We agree that descriptor predictability alone is insufficient evidence of chemically meaningful steering and that decoded molecular changes must be examined directly. We therefore performed the following additional analyses and clarifications. Artifacts will also be available in the anonymous github repo from the paper: <https://anonymous.4open.science/r/MolecuLA-7A68/> (<https://anonymous.4open.science/r/MolecuLA-7A68/>)

- **cLogP and molecular-size coupling.** We agree that the raw cLogP direction is partially coupled to molecular size. At the positive traversal boundary, the raw direction produces $\Delta\text{MolWt} = +18.54$, $\Delta C = +1.28$, $\Delta\text{HAC} = +1.32$, and $\Delta\text{bonds} = +1.42$. However, traversal along the residual cLogP direction substantially reduces this positive-side size growth to $\Delta\text{MolWt} = +4.41$, $\Delta C = +0.14$, $\Delta\text{HAC} = -0.22$, and $\Delta\text{bonds} = -0.48$.

Despite this reduction, the residual direction retains strongly monotonic decoded cLogP ordering, with a median per-seed Spearman correlation of $\rho = 0.959$ and the expected slope direction for all seeds. We therefore do not describe cLogP as a size-independent or disentangled factor. Instead, the evidence supports a more precise conclusion: the raw cLogP direction is partially size-coupled, while the residual direction retains strong decoded cLogP ordering with substantially weaker positive-side size growth. We consequently describe cLogP as a correlated chemical-change axis. The residual-traversal and graph-change analyses are available in the anonymous repository accompanying the paper:

`rebuttal/E04_residual_traversal/`

`rebuttal/E05_chemical_changes/`

- **Expanded graph-level and functional-group analysis.** We expanded the analysis beyond the original SELFIES-level statistics. Across 28,856 distinct cached seed and generated structures, we now measure molecular weight, carbon and heteroatom counts, HeavyAtomCount, ring and aromatic-ring counts, bond counts, branch points, aromatic fractions, Bemis--Murcko scaffold retention, seed similarity, and all 85 RDKit fragment descriptors.

We use these variables to characterise the mechanisms through which decoded chemical change occurs. We do not automatically classify every chemical mediator as a representation confound: functional groups directly define properties such as HBA and strongly determine TPSA, while molecular size is inherent to HeavyAtomCount. The purpose of this analysis is therefore to distinguish sequence-level shortcuts from the concrete chemical changes induced by traversal, rather than to remove all chemically meaningful correlations.

Full results are provided in:

`rebuttal/E05_chemical_changes/`

- **Raw versus residual traversal directions.** We agree that this distinction required direct empirical evaluation. We therefore traversed both the raw probe direction and the corresponding residual direction:

Property	Raw direction ρ	Residual direction: raw property ρ	Residual direction: residual property ρ
FractionCSP3	0.938	0.912	0.893
HBA	0.955	0.929	0.881
TPSA	0.930	0.915	0.865
cLogP	0.935	0.959	0.926
BertzCT	0.944	0.689	0.717
HeavyAtomCount	0.973	0.661	0.746

Raw traversal remains the primary evaluation because it directly targets the chemically defined descriptor. Residual traversal is used as an auxiliary diagnostic showing that the confound-screened signal can itself induce systematic decoded changes.

Importantly, residualising the target against measured confounds does not mathematically impose Euclidean orthogonality between independently fitted latent coefficient vectors. We have revised the discussion to separate statistical residualisation from geometric disentanglement and do not claim that the residual directions are orthogonal latent factors. The complete analysis is available in:

[rebuttal/E04_residual_traversal/](#)

- **Interpretation of the MLP analysis.** We agree that nonlinear MLP decodability alone does not prove steering. The MLP is used only to diagnose whether a property is organised globally and approximately linearly or is more strongly represented through nonlinear local structure. The operational steering test is decoded molecular traversal.

For the raw decoded directions, the median per-seed correlations range from 0.930 to 0.973, with the expected slope sign for all seeds. A molecular-fingerprint predictor or fine-tuned property predictor would evaluate descriptor prediction, but would not test whether displacement along a frozen generative latent direction induces controlled decoded molecular changes. This is precisely the distinction made in the paper between latent readability and latent actionability.

- **Seed-to-lead optimisation and design relevance.** MolecuLA is a representation-analysis study rather than an oracle-guided seed-to-lead optimiser. Consequently, a direct leaderboard comparison against methods such as CLASMO, GB-GA, or MARS would evaluate a substantially different problem involving iterative oracle queries and optimisation-specific search procedures. Nevertheless, to address the reviewer's design concern constructively, we will add a constrained-design evaluation using the existing decoded traversal candidates in the final version. We will report the best property improvement under fingerprint-similarity constraints of $\text{Tanimoto} \geq 0.50$ and $\text{Tanimoto} \geq 0.70$, as well as under Bemis–Murcko scaffold retention. We will additionally report success rate, median property improvement, retained similarity, validity, and the exact candidate and oracle budget.

This analysis is intended as a design-relevant characterisation of the existing traversal procedure, rather than as a claim that MolecuLA is a competitive oracle-guided optimisation algorithm.

- **External steering comparison.** We agree that a task-aligned external comparison is valuable. Comparisons against encoder-only property predictors would not test decoded latent controllability, so we selected external methods that operate through molecular latent-space manipulation. The comparison protocol uses matched held-out seeds, properties, and candidate budgets. Where methodologically appropriate, it also matches latent displacement or retained seed similarity. We evaluate decoded property change, monotonicity, validity, changed-from-seed rate, seed similarity, scaffold retention, constrained-design success and computational cost. The selected external methods are ChemSpacE and ChemFlow. The ChemSpacE comparison is currently being orchestrated under this matched protocol. The completed results, tables, and reproducibility artifacts will be posted during the discussion period in:

[rebuttal/E09_chemspace/](#)

The ChemFlow comparison is more involved and will follow in the final version of the paper.

- **Choice of SELFIES, representation multiplicity, and matched SMILES comparison.** SELFIES was not selected under the assumption that it is universally superior to SMILES. It was selected because decoded molecular validity is central to the question studied in this work. A latent direction is not

operationally actionable if moving along it produces an apparent numerical trend but the resulting strings cannot be decoded into valid molecules.

SELFIES is designed so that valid SELFIES sequences decode through chemically constrained production rules, substantially reducing the syntactic-invalidity failure mode associated with unconstrained molecular string generation. In our evaluation, traversal success is assessed only after the generated representation has been decoded into a valid molecular structure and its molecular properties have been computed. The reported property ordering is therefore measured on decoded molecules rather than inferred only from latent predictions or token-level changes. We also clarify that each canonical molecule in our dataset is converted into one deterministic SELFIES representation. Randomised or alternative SELFIES strings for the same molecule are not mixed during training or evaluation. Therefore, the reported latent organisation cannot be attributed to the model observing multiple alternative SELFIES encodings of each training molecule.

To test representation dependence directly, we completed a matched SMILES VAE experiment using the same dataset and split, architecture, latent dimensionality, objective, training budget, probe protocol, and decoded-traversal evaluation. We also constructed the analogous SMILES-level confound panel.

The matched SMILES model retains latent property organisation and traversal behaviour, but this organisation does not translate into usable molecular control because decoded molecular validity collapses during traversal. This result separates latent descriptor ordering from operational decoded controllability and provides an empirical justification for using SELFIES in the main study. The experiment itself is complete, and we are currently polishing the tables, figures, and reproducibility artifacts. The full results will be posted later today and summarised in a discussion update. They will be available in:

[rebuttal/E08_smiles_representation/](#)

Smiles experiment/ Why SELFIES matter:

Official Comment

by Authors ( Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

 31 Jul 2026, 11:12

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Matched SMILES representation

We completed a matched SMILES VAE experiment using the same dataset and split, architecture, latent dimensionality, objective, training budget, probe protocol, and decoded-traversal evaluation. The model achieved (97.65%) reconstruction accuracy and learned substantial residual descriptor information:

Property	Residual (R^2)	Property	Residual (R^2)
MolWt	0.6270	HeavyAtomCount	0.6091
cLogP	0.6921	TPSA	0.7158
HBD	0.6970	HBA	0.6963
NumRotatableBonds	0.5650	RingCount	0.7017
AromaticRingCount	0.7832	FractionCSP3	0.7837
NumSpiroAtoms	0.1174	NumBridgeheadAtoms	0.1254
BertzCT	0.7504	QED	0.3688
SA-score	0.6947		

The main descriptors are strongly recoverable from the SMILES latent space, and the corresponding directions produce the expected monotonic trends among successfully decoded outputs. However, traversal validity falls to (9.33%), compared with (100%) for SELFIES. Most SMILES traversal outputs are therefore invalid despite high reconstruction accuracy, strong probe (R^2), and correct trends among the small valid subset.

This directly illustrates the distinction between latent readability and decoded actionability. High reconstruction accuracy does not guarantee that latent displacement remains within a region mapped to valid molecular strings, and high (R^2) or monotonic trends do not constitute usable molecular control when more than (90%) of decoded outputs are invalid.

SELFIES was selected not because it is universally superior to SMILES, but because validity is central to the steering question studied here. This result is specific to the matched architecture and protocol; it does not imply that SMILES models are generally incapable of valid steering.

Polished artifacts are available in: [rebuttal/E08_smiles_representation/](#) .

➡ *Replying to Smiles experiment/ Why SELFIES matter:*

External Steering/Baseline: ChempacE

Official Comment

by Authors (👁 Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

📅 31 Jul 2026, 11:31

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

As requested by reviewer up8r, we made a comparison with ChempacE.

It the most directly comparable external baseline for MolecuLA (our model) since both methods operate post hoc on a frozen molecular generator, use property labels only after generative-model training, infer one constant global latent direction per property, traverse a straight line $z+ad$, decode every latent point and evaluate the resulting molecules.

ChemSpacE's published extreme-class linear-SVM direction and our continuous linear probe were fitted in the same frozen representation and evaluated using the same decoder, encoded seeds, candidate budget, unit- L_2 direction normalization, and latent displacement.

Both methods produce property-directed traversal, confirming that the frozen representation supports an independently proposed external steering procedure. Our continuous direction has higher median Spearman monotonicity for five of six properties, while ChemSpacE is slightly higher for cLogP:

Property	MolecuLA median ρ	ChemSpacE median ρ
FractionCSP3	0.938	0.919
HBA	0.955	0.916
TPSA	0.936	0.837
cLogP	0.935	0.949
BertzCT	0.949	0.726
HeavyAtomCount	0.968	0.832

Importantly, the larger ChemSpacE slopes frequently coincide with decoder saturation rather than controlled molecular change. Between 47.6 and 87.4 of ChemSpacE candidates reach the 154-token decoding ceiling, compared with 9.5–22.4 for our directions. This produces extreme decoded-property changes, including approximately +150 HBA, +1409 TPSA, +57

cLogP, and +129.5 heavy atoms at positive endpoints. Although many of these structures remain RDKit-valid, they occur in low-similarity, low-diversity, and frequently scaffold-unevaluable regimes.

To assess whether large property changes are accompanied by chemically reasonable structures, we also report synthetic-accessibility:

The **SA score** is an RDKit heuristic that estimates synthetic difficulty from fragment rarity and structural complexity; **lower values indicate molecules that are heuristically easier to synthesize**, while higher values indicate increasing synthetic difficulty. We therefore report the change relative to the seed, $\Delta SA = SA(x') - SA(x_0)$, where a positive value denotes deterioration. Across all six properties, ChemSpaceE produces larger median synthetic-accessibility penalties:

Property	MolecuLA ΔSA	ChemSpaceE ΔSA
FractionCSP3	+2.424	+3.067
HBA	+2.050	+2.907
TPSA	+1.813	+2.780
cLogP	+1.847	+3.067
BertzCT	+1.635	+4.481
HeavyAtomCount	+2.704	+5.252

Validity, uniqueness, and structural locality

The two methods behave very differently under the same wide matched-displacement traversal:

Property	MolecuLA validity	ChemSpaceE validity	MolecuLA unique/all	ChemSpaceE unique/all	MolecuLA median seed similarity	ChemSpaceE median seed similarity
FractionCSP3	1.000	1.000	0.883	0.291	0.080	0.022
HBA	1.000	0.997	0.749	0.349	0.084	0.029
TPSA	1.000	1.000	0.810	0.449	0.090	0.036
cLogP	1.000	1.000	0.792	0.261	0.092	0.027
BertzCT	1.000	0.919	0.694	0.167	0.082	0.013
HeavyAtomCount	0.970	0.563	0.656	0.301	0.086	0.039

The full $[-150, 150]$ interval is intentionally a stress traversal, so both methods eventually move far from the seed. Nevertheless, at exactly matched Euclidean latent displacement, MolecuLA retains substantially higher canonical diversity and seed similarity for every property.

ChemSpaceE's sharper movement often converges toward a much smaller set of decoder outputs. At the two endpoints, it produces only 2 distinct cLogP structures, 5 HBA structures, and 6 BertzCT structures among approximately 100 candidate occurrences, whereas MolecuLA retains between 59 and 94 distinct endpoint structures for five properties. These results show that large decoded property changes can coexist with strong convergence toward decoder attractors.

The complete method-equivalence audit, released-protocol transfer, strict-success evaluation, signed- quality curves, scaffold statistics, SA/NP analysis, decoder-saturation results, and per-property constrained-design tables are available in: [rebuttal/E09_chemspace/](#).

Seed-to-lead optimisation and design relevance

Official Comment

by Authors ( Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

 31 Jul 2026, 11:43

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Our MolecuLA is a representation-analysis framework rather than an iterative, oracle-guided seed-to-lead optimiser. Nevertheless, we added a fixed-budget, seed-centred constrained-design evaluation to determine whether the learned directions produce practically usable candidates before the decoded molecule loses structural similarity to its starting point.

For each property, seed, method, and similarity cutoff, a traversal is considered successful only if it contains at least one candidate satisfying all of the following:

1. nonzero traversal coordinate, $\alpha \neq 0$;
2. RDKit-valid decoding;
3. canonical structure different from the seed;
4. strict improvement in the requested property direction; and
5. radius-2 Morgan fingerprint similarity to the seed above the specified threshold.

We use the similarity cutoffs $\tau \in 0.2, 0.4, 0.6$ reported in the ChemSpaceE evaluation. For property maximisation, the success condition is

$$\exists, x_\alpha : \quad \alpha \neq 0, \quad x_\alpha \neq x_0, \quad f(x_\alpha) > f(x_0), \quad T!(x_\alpha, x_0) \geq \tau,$$

with the property inequality reversed for minimisation.

We first used the exact 21-candidate budget of the published ChemSpaceE traversal. Because this corresponds to 15 latent units between neighbouring points over the wide interval $[-150, 150]$, it undersamples ChemSpaceE's narrow transition between reconstruction and structurally distant outputs. We therefore additionally evaluated a denser 100-candidate sensitivity using the same seeds, directions, decoder, and traversal range.

Minimum seed similarity	ChemSpaceE mean success	MolecuLA mean success
0.2	0.563	0.917
0.4	0.223	0.600
0.6	0.080	0.230

The denser evaluation restores ChemSpaceE success in 17 of the 18 property-by-threshold settings, confirming that the zeroes observed with the coarse 21-point grid were partly caused by sampling resolution. Nevertheless, MolecuLA achieves higher constrained success for **all six properties at all three nonzero similarity cutoffs**.

At similarity 0.2, MolecuLA produces an improving candidate for 86--98 of seeds, compared with 30--68 for ChemSpaceE. At similarity 0.4, the corresponding ranges are 54--66 versus 4--42, while at similarity 0.6 they are 14--30 versus 0--18.

At zero similarity constraint, both methods achieve essentially perfect maximisation success. This shows why unconstrained property improvement alone is not informative: ChemSpaceE frequently reaches extreme property values only after moving into low-similarity, low-diversity, token-saturated decoder regions. MolecuLA's continuous directions move more gradually and therefore provide substantially more intermediate candidates that retain structural similarity to the seed.

The dense comparison evaluates $50 \times 100 \times 6 = 30,000$ candidates per method, or 60,000 candidates in total. The matched 21-point experiment evaluates 6,300 candidates per method. No adaptive search, reinforcement-learning loop, or repeated direction optimisation is used; the candidate sets are produced by direct traversal of the fixed learned directions.

For every successful seed, the complete analysis additionally reports the achieved property improvement, retained fingerprint similarity, validity, scaffold evaluability and retention, and synthetic-accessibility change. Bemis--Murcko scaffold statistics are reported as supplementary diagnostics rather than the primary constraint because scaffold retention is undefined when either the seed or candidate is acyclic. Fingerprint similarity provides a graded constraint applicable to all molecules.

We also evaluated a stricter sensitivity requiring

$$SA(x_\alpha) \leq SA(x_0),$$

in addition to property improvement and fingerprint similarity. MolecuLA is never worse under this constraint and performs better for five of the six properties at similarity 0.2, with equality for BertzCT. We treat this only as supplementary evidence because the SA score is a heuristic rather than a synthesis oracle.

These results should not be interpreted as claiming that MolecuLA is a competitive general-purpose seed-to-lead optimiser. Rather, they provide a design-relevant characterisation of the learned representation: MolecuLA's continuous directions yield a graded sequence of valid, property-improving candidates and substantially broader seed coverage under explicit structural constraints than the sharper ChemSpace extreme-class directions.

Additional insight is available at [rebuttal/E09_chemspace/Constrained optimization](#) . and [rebuttal/E06_traversal_quality/](#)

➡ Replying to Seed-to-lead optimisation and design relevance

Official Comment by Reviewer up8r

Official Comment by Reviewer up8r 📅 01 Aug 2026, 08:58

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Thanks for your reply. However, I think the authors don't understand my concerns at all.

1. The author failed to articulate the storyline, the motivation of this paper is unclear. Take an example, if you want to use LLM to learn chemical space, why don't you use transformer directly? I didn't see the necessity of an extra VAE model. I believe a simple SFT on those properties can also do the same thing. I'm asking for extra baselines, such as UniMol, MolRAE, and Suiren. Those model are designed for molecule representation, can be directly compared with your method. If you're talking about language modelling approach, why don't you compare with ChemBERTa or MolGPT? The baseline you choose is just another weak model.
2. The selection of SELFIES. SELFIES is claimed to address the validity issue, however, at the cost of redundancy representation. Under your application, I don't think the slight error of SMILES could be a problem. Besides, if you ever use SELFIES for real application, you can easily found SELFIES will failed when atom number is increasing to 100. This is also attribute to the redundant decoding mechanism.
3. Linear probing is an out-dated method. If you want to do properties prediction, just add another head and SFT. The latent space of modern deep learning models are likely incompatible with ML models. I didn't see any insight from your LP experiments while I appreciate the workload. Also, the properties you choose is just some toy case. I don't know why predict the atom numbers can appeared in a NeruIPS submitted paper. It's so trivial.

Overall, I strongly against the publication of this paper. The method is simple combination of out-dated model. The motivation and insight is so weak. The baseline is (only one) so narrow and the choice is also making no sense.

Addressing the concerns

Official Comment

by Authors ( Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

 01 Aug 2026, 17:52 (modified: 01 Aug 2026, 22:00)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

 Revisions (/revisions?id=qy7SaIoeu4)

Comment:

We thank reviewer **up8r** for the follow-up. We believe the latest comment evaluates the work primarily as a supervised property-prediction study, whereas the paper asks a different question: whether property-blind generative training produces latent information that is linearly readable, survives measured shortcut controls, and forms an actionable direction under decoded molecular traversal. We clarify the three points below.

1. Motivation, VAE design, SFT, and external baselines

The paper does not propose using an LLM to learn chemical space; it studies an unsupervised Transformer-VAE. The model already uses a Transformer, while the VAE supplies the continuous bottleneck required to define a trajectory $z + \alpha d$ and decode every displaced latent point. Removing it would remove the geometric object being analysed.

Supervised fine-tuning would answer a different question. It would inject the target properties into the model and test whether it can be trained for prediction or optimisation. Our question is whether those directions emerge before property supervision, with the generator frozen.

Encoder-oriented representation models such as Uni-Mol, or supervised property predictors such as ChemBERTa, are therefore not direct baselines for decoded latent traversal. A comparable method must expose a continuous generative latent space, a post-hoc global direction, and a decoder applicable after displacement. ChemSpacE was selected because it satisfies these requirements and was explicitly identified as an appropriate external steering baseline. We evaluated it using the same frozen representation, decoder, seeds, candidate budget, direction normalisation, and displacement. The latest comment identifies no error in this comparison and only characterises the baseline as weak.

2. SELFIES selection and representation redundancy

SELFIES non-uniqueness does not create the claimed ambiguity in our protocol. Each canonical molecule is converted into one deterministic SELFIES sequence, and randomised alternatives are not mixed during training. Generated sequences are decoded and canonicalised before validity, uniqueness, novelty, similarity, or scaffold statistics are computed. Different SELFIES strings representing the same graph therefore collapse to the same canonical molecule.

Moreover, the matched SMILES result is not a slight degradation. The SMILES model achieved 97.65 reconstruction, strong residual R^2 , and the expected trends among successfully decoded outputs, but only 9.33% traversal validity, compared with 100% for SELFIES. Thus, more than 90% of displaced SMILES latent points did not yield valid molecules. This directly motivates SELFIES for a study in which decoded molecular actionability, rather than prediction alone, is the operational criterion. We make no universal claim about SELFIES or molecules outside the studied size regime.

3. Linear probing and the selected descriptors

Linear probing is not presented as a state-of-the-art property predictor. It is deliberately used because its coefficient vector defines one constant global direction whose decoded actionability can be tested. A nonlinear predictor may recover more information, but it does not establish that the property is organised along one globally traversable axis. The paper therefore uses linear probes for candidate directions, MLP probes for nonlinear recoverability, and decoded traversal as the operational test.

Adding a supervised head and fine-tuning would alter the representation and could not determine whether the structure already existed after property-blind training. Neural latent vectors are also not incompatible with classical estimators; fitting restricted probes to frozen neural representations is precisely what allows their geometry to be analysed without changing them.

Finally, HeavyAtomCount is not included because atom counting is a difficult prediction task. It is an explicit size-sensitive control used to reveal how a genuine but chemically entangled size direction behaves and to distinguish it from representation-level token shortcuts.

Overall, we appreciate that the reviewer may prefer supervised representation or property-prediction studies. However, those methods answer a different question. Our objective is to determine whether property-blind generative training produces globally readable, decoded-actionable latent directions that survive explicit shortcut controls. This requires a continuous generative bottleneck, restricted probes, and decoded traversal. The residual-direction experiments, nonlinear confound audit, graph-change and functional-group analyses, matched SMILES study, ChemSpace comparison, and constrained-design evaluation directly address the technical concerns raised in the original review. The latest comment identifies no error in these analyses, but instead argues for replacing the study with a supervised prediction task.

Official Review of Submission24784 by Reviewer aat6

Official Review by Reviewer aat6 📅 22 Jun 2026, 06:08 (modified: 04 Aug 2026, 15:46)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer aat6

📄 Revisions (/revisions?id=dfs6GMqEoc)

Summary:

The paper trains an unsupervised SELFIES-based Transformer-VAE and investigates whether molecular properties are organized as meaningful latent directions. After training, the model is frozen and linear probes are fitted to RDKit descriptors, with probe weights used as candidate latent steering directions. The main contribution is a confound-aware evaluation framework that distinguishes genuine chemical organization from sequence-level shortcuts. The framework uses SELFIES-derived confounds, residualized probing, and decoded-molecule validation to determine whether property predictability reflects chemistry rather than representation artifacts.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths

1. The paper correctly identifies that high probe R^2 can reflect SELFIES sequence-level shortcuts rather than chemical organization, and insists on decoded-molecule traversal as the real test of a steering direction.
2. The robustness analysis is thorough that goes beyond single-run reporting. Bootstrap direction stability, permutation control, etc. show that the recovered directions are reproducible and not chance artifacts.
3. Reproducibility and transparency are excellent. Full hyperparameters, seeds, dataset license, and an anonymized repository are provided, and the limitations section honestly states the scope constraints.

Weaknesses

1. The contribution is an evaluation and analysis framework. The individual components, such as Transformer-VAE, linear probes, residualization, latent traversals, MLP probing, are all established. At NeurIPS, we usually expect a novel generative or steering method.
2. All experiments use a single SELFIES dataset, a single model family, and a single molecular representation. It is unclear whether the conclusions transfer to graph-based generative models, diffusion models, pretrained molecular language models, or SMILES-based architectures.
3. The confound panel contains only length, branch-token count, ring-token count, and token entropy, which is unlikely to capture all sources of shortcut learning. More fundamentally, residualization uses a linear/Ridge confound predictor, so it only removes the linearly predictable confound component; Appendix D.2 itself shows residual directions still retain non-negligible cosine alignment with the confound subspace. Therefore, I think the method under-corrects by design, not just because the panel is short.

- Latent directions are evaluated only against the authors' own non-autoregressive ablations, with no comparison to other molecular optimization or steering methods (e.g., ChemFlow, ChemSpace). This makes practical impact for molecular design hard to assess.
- The study covers only RDKit descriptors. The conclusions may not extend to more challenging objectives such as bioactivity, toxicity, docking, pharmacokinetics, or synthesis outcomes.

Quality: 4: excellent

Clarity: 3: good

Significance: 2: not good

Originality: 2: not good

Questions:

- How far do the conclusions extend beyond a single SELFIES Transformer-VAE? Can you test if the framework transfers to SMILES-based, graph-based, or diffusion-based molecular models, or to a second dataset?
- How sensitive are the residualized conclusions to the chosen confound panel? Would adding scaffold statistics, graph-size measures, or functional-group counts materially change which descriptors survive as "chemically meaningful"?
- Did you evaluate nonlinear confound models, and how much confound signal might a nonlinear confounder remove that the current procedure leaves behind?
- For highly confounded but still-traversable properties such as HeavyAtomCount and BertzCT, what concrete chemical changes occur along the traversal? Specifically, what evidence distinguishes genuine topological steering from a SELFIES-length/token-count effect?
- Can you report traversal validity, uniqueness, and novelty as a function of traversal magnitude (α)?
- Can you compare against at least one external steering/optimization method you already cite (e.g., ChemFlow or ChemSpace), and show if your method recovers the same robust-vs-confounded descriptor split or steer the clean descriptors (cLogP, TPSA, HBA, FractionCSP3) more reliably?

Limitations:

Yes.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

None.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

Rebuttal by Authors

Rebuttal

by Authors ( Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

 28 Jul 2026, 13:59 (modified: 28 Jul 2026, 16:08)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

 Revisions (/revisions?id=euDnkVlhT4)

Rebuttal:

We thank reviewer **aat6** for recognising the paper's distinction between probe predictability and decoded steering, its robustness analysis, and its reproducibility. We agree that the original submission left questions regarding generality, nonlinear confounding, concrete decoded changes, traversal quality, and external comparison. We therefore performed the following analyses and clarified the intended contribution.

Artifacts are available in the anonymous repository: <https://anonymous.4open.science/r/MolecuLA-7A68/> (<https://anonymous.4open.science/r/MolecuLA-7A68/>)

Nature of the contribution

MolecuLA does not introduce a new molecular generator or optimisation algorithm. Its contribution is an evaluation framework distinguishing: (i) information readable from a molecular latent space, (ii) information that remains after controlling for measured representation-level shortcuts, and (iii)

information that forms an actionable direction under decoded molecular traversal.

The additional experiments broaden this evaluation through nonlinear nuisance modelling, residual-direction traversal, graph-change decomposition, traversal-quality curves, cross-dataset transfer, independent replication, a matched SMILES representation, and an external steering comparison. We will revise the paper to state this contribution more explicitly.

Generality beyond one dataset and one representation

We performed two complementary experiments on ZINC250K.

Property	Frozen original model	Retrained ZINC model
FractionCSP3	0.8875	0.8541
HBA	0.6579	0.5800
TPSA	0.6326	0.5952
cLogP	0.6020	0.6576
BertzCT	0.7753	0.7523
HeavyAtomCount	0.7260	0.6872

First, we froze the original MolecuLA encoder and fitted new ZINC250K probes. Held-out R^2 values range from 0.6020 to 0.8875, showing that the frozen encoder retains substantial linearly accessible chemical information under dataset shift. This is cross-dataset representation transfer with post-hoc probe refitting, not zero-shot transfer of the original coefficients. The ZINC-specific directions also produced the expected monotonic decoded property changes through the original frozen decoder.

Second, we independently retrained the same architecture on ZINC250K. Held-out R^2 values range from 0.5800 to 0.8541, and decoded traversals again move the corresponding properties monotonically in the expected direction. The first experiment demonstrates cross-dataset reuse of the representation; the second demonstrates replication on a new dataset. Together, they show that the central result is not specific to one dataset instance or training run.

Full protocols, probe results, and traversal plots are in `rebutta1/E07_Zinc250k_second_dataset`.

Matched SMILES representation

We completed a matched SMILES VAE experiment using the same dataset and split, architecture, latent dimensionality, objective, training budget, probe protocol, and decoded-traversal evaluation. The model achieved 97.65 reconstruction and learned substantial residual descriptor information:

Property	Residual R^2	Property	Residual R^2
MolWt	0.6270	HeavyAtomCount	0.6091
cLogP	0.6921	TPSA	0.7158
HBD	0.6970	HBA	0.6963
NumRotatableBonds	0.5650	RingCount	0.7017
AromaticRingCount	0.7832	FractionCSP3	0.7837
NumSpiroAtoms	0.1174	NumBridgeheadAtoms	0.1254
BertzCT	0.7504	QED	0.3688
SA-score	0.6947		

The main descriptors are strongly recoverable from the SMILES latent space, and the corresponding directions produce the expected monotonic trends among successfully decoded outputs. However, traversal validity falls to 9.33, compared with 100 for SELFIES. Most SMILES traversal outputs are therefore invalid despite high reconstruction, strong probe R^2 , and correct trends among the small valid subset.

This directly illustrates the distinction between latent readability and decoded actionability. High reconstruction does not guarantee that latent displacement remains in a region mapped to valid molecular strings, and high R^2 or monotonic trends do not constitute usable molecular control when more than 90 of outputs are invalid.

SELFIES was selected not because it is universally superior to SMILES, but because validity is central to the steering question studied here. This result is specific to the matched architecture and protocol; it does not imply that SMILES models are generally incapable of valid steering. Polished artifacts will be posted during discussion in [rebuttal/E08_smiles_representation/](#).

Nonlinear confounding

We repeated residualisation with nonlinear nuisance models under a leakage-controlled protocol: five-fold cross-fitted nuisance predictions on training, with validation and test residuals generated only by a model trained on the training partition.

Property	Nonlinear-residual latent R^2
FractionCSP3	0.761
HBA	0.662
TPSA	0.641
cLogP	0.641
BertzCT	0.648
HeavyAtomCount	0.563

Fresh nonlinear models were then trained to predict these residuals from the original confound panel on held-out data. The maximum remaining predictability was $R^2 = 0.0091$ for HeavyAtomCount; RingCount, ExactMolWt, and MolWt yielded 0.0017, 0.0010, and 0.0009, respectively, with nearly all other values below 0.001.

Thus, residual targets remain predictable from the latent space, with R^2 from 0.563 to 0.761, while being almost completely unpredictable from the measured confounds. The surviving signal is therefore not explained by underfitting of the original linear nuisance relation. This conclusion remains limited to the specified confound panel. Full results are in [rebuttal/E03_nonlinear_confounds/](#).

Residual-direction alignment with the confound subspace

Residualisation removes the target component statistically predictable from measured confounds; it does not impose Euclidean orthogonality between independently fitted latent coefficient vectors. Non-zero cosine alignment between a residual probe and a confound direction therefore does not by itself show that the residual target remains predictable from those confounds.

The held-out nonlinear audit is the relevant test: residual properties remain strongly predictable from latent representations, whereas their remaining predictability from confounds is at most $R^2 = 0.0091$. We will revise the paper to distinguish statistical residualisation from geometric orthogonalisation and do not claim disentangled or orthogonal latent factors.

Concrete chemical changes for HeavyAtomCount and BertzCT

For HeavyAtomCount, the positive traversal boundary produces $\Delta\text{MolWt} = +20.50$, $\Delta C = +1.24$, $\Delta\text{HeavyAtomCount} = +1.46$, $\Delta\text{bonds} = +1.56$, and $\Delta\text{branch points} = +0.66$. We therefore describe HeavyAtomCount as a molecular-size steering direction, not an independent or disentangled factor.

For BertzCT, full-range decoded correlations are 0.997 with MolWt, 0.995 with HeavyAtomCount, 0.998 with bond count, and 0.765 with branch points. BertzCT is therefore a correlated size/topology axis rather than an isolated aromatic-complexity direction.

Across 28,856 distinct cached seed and generated structures, we additionally measure carbon and heteroatom counts, ring and aromatic-ring counts, aromatic fractions, scaffold retention, seed similarity, and all 85 RDKit fragment descriptors. Full results are in [rebuttal/E05_chemical_changes/](#).

Traversal quality versus magnitude

We report traversal quality for every signed magnitude across 1, 300 property/direction/ α groups: validity, unique canonical fraction, changed-from-seed rate, median and interquartile-range seed similarity, Bemis--Murcko scaffold retention, exact training-set novelty, and nearest-training fingerprint similarity where available.

Rather than infer a single automatic "trust boundary," we present signed- α curves and fixed seed-similarity and scaffold-retention constraints. Directional ordering persists broadly, while sufficiently large displacement reduces seed similarity and scaffold retention. We therefore distinguish global latent organisation from local scaffold-preserving control. Full results are in [rebuttal/E06_traversal_quality/](#).

External steering comparison

The ChemSpaceE comparison is running under a matched protocol using the same held-out seeds, properties, and candidate budget, with latent displacement or retained seed similarity matched where appropriate. We evaluate decoded property change, monotonicity, validity, changed-from-seed rate, seed similarity, scaffold retention, constrained-design success, computational cost, and oracle calls. Results and artifacts will be posted during discussion in [rebuttal/E09_chemspace/](#). ChemFlow requires a more involved learned-vector-field implementation and will be included in the final version.

Objectives beyond RDKit descriptors

The present results do not establish steering for experimental bioactivity, toxicity, docking, pharmacokinetics, or synthesis outcomes. These require different supervision, evaluation protocols, and often external predictive or experimental oracles. We have clarified this limitation in the paper's conclusions. Extending the framework to such objectives is important future work, but extensive new docking or bioactivity pipelines would constitute a substantially different study.

Response-1 to authors' rebuttal

Official Comment by Reviewer aat6 31 Jul 2026, 07:09

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

While I acknowledge that the authors managed to comprehensively address some of the weaknesses I pointed out, unfortunately the following weaknesses remain and anchor my current score.

Weaknesses the rebuttal does not change

The contribution remains an evaluation and analysis framework assembled from established components rather than a new generative or steering method (W1), yet the study remains confined to computed RDKit descriptors, with experimental objectives such as bioactivity, toxicity, docking, and pharmacokinetics out of scope (W5). These are legitimate scoping choices, but they cap the significance and originality at borderline for this venue, which is why the steering-validation and external-baseline gaps are decisive for my current score.

What could still change my score

I would move to a borderline accept if the authors show "both" of the following during the ongoing discussion period:

1. Residualized-direction (or confound-orthogonalized) traversals that steer the clean descriptors (cLogP, TPSA, HBA, FractionCSP3) with monotonic decoded behavior comparable to the raw-direction traversals. They must demonstrate that steering is not an artifact of the un-corrected directions.
2. The ChemSpaceE comparison under the matched protocol they describe, showing where their diagnostic directions stand relative to an external method. I would not require the expanded confound panel or the harder biological objectives for a positive score, as those are reasonable future work.

Residualized direction steering

Official Comment

by Authors ( Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

 31 Jul 2026, 08:33 (modified: 31 Jul 2026, 08:57)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

 Revisions (/revisions?id=Y0nhlOV2mJ)

Comment:

The residualized-direction analysis was planned earlier during the rebuttal, but we inadvertently included it only in our response to reviewer **up8r**. We apologize for this oversight. The complete analysis is provided below.

1. Raw versus residual traversal directions

We agree that this distinction required direct empirical evaluation. We therefore traversed both the raw probe direction and the corresponding residual direction:

Property	Raw direction (ρ)	Residual direction: raw property (ρ)	Residual direction: residual property (ρ)
FractionCSP3	0.938	0.912	0.893
HBA	0.955	0.929	0.881
TPSA	0.930	0.915	0.865
cLogP	0.935	0.959	0.926
BertzCT	0.944	0.689	0.717
HeavyAtomCount	0.973	0.661	0.746

Here, ρ is computed separately for each of the 50 held-out seed molecules as the Spearman rank correlation between the signed traversal coordinate α and the decoded property trajectory; the table reports the median across seeds. Thus, $\rho = 1$ denotes a perfectly monotonic increase across the traversal, whereas $\rho \approx 0$ indicates no consistent ordering.

Residual-direction traversal remains monotonic for all six properties. For the four clean descriptors specifically highlighted in the discussion, residual-direction traversal remains directly comparable to raw traversal. The median correlations for the decoded raw property change only from $0.938 \rightarrow 0.912$ for FractionCSP3, $0.955 \rightarrow 0.929$ for HBA, and $0.930 \rightarrow 0.915$ for TPSA, while cLogP improves from $0.935 \rightarrow 0.959$. The difference in median ρ is therefore at most 0.026, and the fitted slope has the expected sign for all 50 seeds for each of these four properties.

The third column provides an even stricter diagnostic: after traversing the residual direction, we recompute the nuisance-corrected property value of each decoded molecule. These trajectories also remain strongly monotonic, with median $\rho = 0.893$ for FractionCSP3, $\rho = 0.881$ for HBA, $\rho = 0.865$ for TPSA, and $\rho = 0.926$ for cLogP. Consequently, the traversal is not merely changing the raw descriptor through the component already explained by the measured SELFIES confounds; the confound-screened component itself changes systematically after decoding.

Raw traversal remains the primary evaluation because it directly targets the chemically defined descriptor. Residual traversal is used as an auxiliary diagnostic showing that the confound-screened signal can itself induce systematic decoded changes.

Importantly, residualizing the target against measured confounds does not mathematically impose Euclidean orthogonality between independently fitted latent coefficient vectors. We have revised the discussion to distinguish statistical residualization from geometric disentanglement, and we do not claim that the residual directions represent orthogonal latent factors.

The complete analysis is available in: [rebuttal/E04_residual_traversal/](#)

2. ChemSpace comparison

The ChemSpace comparison is still in progress. Because this experiment requires careful validation, we prefer not to report preliminary or incomplete results that could lead to an unreliable conclusion. We will provide the complete analysis once it has been thoroughly validated.

➡ *Replying to Residualized direction steering*

ChemSpace diagnostic 1/2

Official Comment

by Authors (👁 Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

📅 31 Jul 2026, 11:00

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

We have completed the ChemSpace comparison under a controlled shared-backbone protocol. We will call our model MolecuLA for simplicity.

ChemSpace's published extreme-class linear-SVM direction and our continuous linear probe were fitted in the same frozen representation and evaluated using the same decoder, encoded seeds, candidate budget, unit- L_2 direction normalization, and latent displacement.

Both methods produce property-directed traversal, confirming that the frozen representation supports an independently proposed external steering procedure. Our continuous direction has higher median Spearman monotonicity for five of six properties, while ChemSpace is slightly higher for cLogP:

Property	MolecuLA median ρ	ChemSpace median ρ
FractionCSP3	0.938	0.919
HBA	0.955	0.916
TPSA	0.936	0.837
cLogP	0.935	0.949
BertzCT	0.949	0.726
HeavyAtomCount	0.968	0.832

Importantly, the larger ChemSpace slopes frequently coincide with decoder saturation rather than controlled molecular change. Between 47.6 and 87.4 of ChemSpace candidates reach the 154-token decoding ceiling, compared with 9.5–22.4 for our directions. This produces extreme decoded-property changes, including approximately +150 HBA, +1409 TPSA, +57 cLogP, and +129.5 heavy atoms at positive endpoints. Although many of these structures remain RDKit-valid, they occur in low-similarity, low-diversity, and frequently scaffold-unevaluable regimes.

To assess whether large property changes are accompanied by chemically reasonable structures, we also report synthetic-accessibility and natural-product-likeness scores.

The **SA score** is an RDKit heuristic that estimates synthetic difficulty from fragment rarity and structural complexity; **lower values indicate molecules that are heuristically easier to synthesize**, while higher values indicate increasing synthetic difficulty. We therefore report the change relative to the seed, $\Delta SA = SA(x') - SA(x_0)$, where a positive value denotes deterioration.

The **NP-likeness score** measures how closely a molecule resembles structural patterns observed in natural products; **higher values indicate greater natural-product-like character**. It is complementary to SA score and should not be interpreted as a direct

measure of synthesizability, drug-likeness, stability, or biological activity. Chemical-quality diagnostics reinforce this distinction. Across all six properties, ChemSpaceE produces larger median synthetic-accessibility penalties:

Property	MolecuLA Δ SA	ChemSpace Δ SA
FractionCSP3	+2.424	+3.067
HBA	+2.050	+2.907
TPSA	+1.813	+2.780
cLogP	+1.847	+3.067
BertzCT	+1.635	+4.481
HeavyAtomCount	+2.704	+5.252

NP-likeness does not provide a consistent compensating advantage for ChemSpaceE and increases more along our direction for several properties. We interpret NP-likeness only as a complementary fragment-pattern diagnostic, not as evidence of synthesizability.

Validity, uniqueness, and structural locality

The two methods behave very differently under the same wide matched-displacement traversal:

Property	MolecuLA validity	ChemSpaceE validity	MolecuLA unique/all	ChemSpaceE unique/all	MolecuLA median seed similarity	ChemSpaceE median seed similarity
FractionCSP3	1.000	1.000	0.883	0.291	0.080	0.022
HBA	1.000	0.997	0.749	0.349	0.084	0.029
TPSA	1.000	1.000	0.810	0.449	0.090	0.036
cLogP	1.000	1.000	0.792	0.261	0.092	0.027
BertzCT	1.000	0.919	0.694	0.167	0.082	0.013
HeavyAtomCount	0.970	0.563	0.656	0.301	0.086	0.039

The full $[-150, 150]$ interval is intentionally a stress traversal, so both methods eventually move far from the seed. Nevertheless, at exactly matched Euclidean latent displacement, MolecuLA retains substantially higher canonical diversity and seed similarity for every property.

ChemSpaceE's sharper movement often converges toward a much smaller set of decoder outputs. At the two endpoints, it produces only 2 distinct cLogP structures, 5 HBA structures, and 6 BertzCT structures among approximately 100 candidate occurrences, whereas MolecuLA retains between 59 and 94 distinct endpoint structures for five properties. These results show that large decoded property changes can coexist with strong convergence toward decoder attractors.

➔ *Replying to ChemSpace diagnostic 1/2*

ChemSpace diagnostic 2/2

Official Comment

by Authors ( Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

 31 Jul 2026, 11:08

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Similarity-constrained molecular design

We next evaluated a fixed-budget seed-centered design task. A seed is considered successful only if the traversal produces at least one candidate satisfying all of the following:

1. nonzero α ;
2. RDKit validity;
3. canonical structure different from the seed;
4. strict property improvement in the requested direction; and
5. radius-2 Morgan similarity to the seed above the stated threshold.

At zero similarity constraint, both methods achieve nearly perfect success, showing that unconstrained property improvement alone is not discriminative. Once structural proximity is required, the difference is substantial.

The original 21-point traversal is an exact candidate-budget match to the published ChemSpace protocol, but its 15-unit spacing undersamples ChemSpace's narrow local transition. We therefore also report a denser 100-candidate sensitivity using the same seeds and traversal range.

For the 100-candidate comparison:

Minimum seed similarity	ChemSpace mean success	MolecuLA mean success
0.2	0.563	0.917
0.4	0.223	0.600
0.6	0.080	0.230

The denser grid restores ChemSpace success in 17 of the 18 property-by-threshold settings, confirming that the zeroes in the coarse grid were partly caused by sampling resolution. Nevertheless, MolecuLA remains higher for all six properties at all three nonzero similarity thresholds.

At similarity 0.2, MolecuLA succeeds for 86–98 of seeds, compared with 30–68 for ChemSpace. At similarity 0.4, the corresponding ranges are 54–66 versus 4–42, and at similarity 0.6 they are 14–30 versus 0–18.

A stricter sensitivity additionally requires that the candidate does not worsen the seed's SA score. At similarity 0.2, MolecuLA is never worse and is better for five of the six properties, with equality for BertzCT. We treat this only as a supplementary sensitivity because SA is a heuristic rather than a synthesis oracle.

Scaffold behavior

Scaffold retention must be interpreted together with scaffold evaluability. A Bemis–Murcko retention value is undefined when either the seed or candidate has no nonempty scaffold.

Only 7.5–15.4 of valid ChemSpace candidate pairs have a nonempty scaffold on both sides, compared with 62.9–81.7 for MolecuLA. This does not mean that every ChemSpace candidate explicitly removes an existing scaffold; many decoded structures become acyclic. However, it shows that the traversal often leaves the seed's scaffold-bearing chemical regime.

Reporting retention only among evaluable pairs would therefore hide the dominant behavior. We report both scaffold evaluability and retention in the complete analysis.

The results show that the two direction learners summarize different aspects of the same frozen representation:

ChemSpace's extreme-SVM direction provides very fast movement toward extreme-property regions and slightly stronger cLogP rank monotonicity. However, its trajectories enter long-sequence, low-diversity, low-similarity, and synthetically less favorable decoder regimes much earlier.

MolecuLA's continuous probe provides more gradual movement, higher rank monotonicity for five of six properties, substantially greater canonical diversity, much lower decoder saturation, smaller SA-score deterioration for every property, and more property-improving

candidates under all tested fingerprint-similarity constraints.

The complete method-equivalence audit, released-protocol transfer, strict-success evaluation, signed- α quality curves, scaffold statistics, SA/NP analysis, decoder-saturation results, and per-property constrained-design tables are available in: [rebutta1/E09_chemspace/](#) .

➤ *Replying to ChemSpace diagnostic 2/2*

Response to authors' comment

Official Comment by Reviewer aat6 📅 01 Aug 2026, 05:16

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

I have examined the repository, and found that the reported results are consistent with the underlying data. Both conditions I set out are met, and, as a result, I am raising my score to borderline accept.

➤ *Replying to Response to authors' comment*

Official Comment by Authors

Official Comment

by Authors (👁 Attila Cangi (/profile?id=~Attila_Cangi1), Bartosz Brzoza (/profile?id=~Bartosz_Brzoza1), Zakaria Elabid (/profile?id=~Zakaria_Elabid1), Jan Andrzejewski (/profile?id=~Jan_Andrzejewski1))

📅 03 Aug 2026, 16:46

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors

Comment:

Dear Reviewer aat6. Thank you very much for your careful reassessment and for taking the time to examine the repository and verify the reported results. We sincerely appreciate your constructive feedback and your decision to update the score; in the revised manuscript, we will incorporate the additional experiments conducted during the discussion period and more clearly articulate the remaining scope and significance limitations you highlighted.